

|             |                                                                                                  |
|-------------|--------------------------------------------------------------------------------------------------|
| <b>Q1/E</b> | Write a C++ program using the <b>nested for</b> to repeat print numbers from 4 to 1 three times. |
|-------------|--------------------------------------------------------------------------------------------------|

|             |                                                                                                                                                        |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1/D</b> | Write a C++ program to input a variable that can accept any of the following values: A, &, B, 1, *, 3 as input then print the variable type and value? |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                                                               |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1/C</b> | Write a C++ program that inputs a number and determines (يحدد) whether the subtraction of the number from 3 is an odd number. |
|-------------|-------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                                                                               |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Q1/B</b> | Write a C++ program to enter numbers and stop if the summation of interred numbers is greater than <b>80</b> using a <b>while statement</b> . |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                                                             |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|
| <b>Q1/A</b> | Write a C++ program that requests five integer values from the user. It then prints the maximum and minimum values entered. |
|-------------|-----------------------------------------------------------------------------------------------------------------------------|

|             |                                                                                         |
|-------------|-----------------------------------------------------------------------------------------|
| <b>Q1/F</b> | Write a C++ program to accept four numbers and find the product of their reminder by 3. |
|-------------|-----------------------------------------------------------------------------------------|

Student Name:

Student Class:

**Q1/a.** Write a C++ program uses the if...else to test for divisibility (قابلية القسمة) of two integer numbers. **8 marks**

**b.** Write a program to find the value of **Z** from the following (using a switch statement). **7 marks**

$$Z = \begin{cases} x + 5 & x > 0 \\ x - 5 & x \leq 0 \end{cases}$$

**Q2/a.** Circle the correct answer.

**9 marks**

|                                                                                                                                                                        |                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>1.</b> If <b>n=6</b> , what will be the result of:<br>switch(n) {<br>case 3: cout << (n-5)<<endl;<br>case 6: cout << (n+5)<<endl;<br>default: cout << (n*5)<<endl;} | a) 11    b) 30<br>30        11<br><br>c)11        d) 30 |
| <b>2.</b> What will be the result of:<br>int n=6;<br>cout<< (3n);                                                                                                      | a) error    b) 36<br>c) 18        d) 63                 |
| <b>3.</b> Is this code correct?<br>char A='R';<br>cout<<R;                                                                                                             | a) Yes<br>b) No                                         |

**b.** Evaluate the following expressions if y=2:

**6 marks**

a)  $Z = 3 * 8 / 2 * (2 * y + 5 / 2);$

b)  $Z = 7 * y \% 2 + 2 * (3 + (y \% 3 + 2));$

c)  $y = ((2 == 4) \&\& (7 > 5));$

Student Name:

Student Class:

**Q1/a.** Write a C++ program uses the if....else to test for divisibility (قابلية القسمة) of two integer numbers. **8 marks**

**b.** Write a program to find the value of **Z** from the following (using a switch statement). **7 marks**

$$Z = \begin{cases} x + 5 & x > 0 \\ x - 5 & x \leq 0 \end{cases}$$

**Q2/a.** Circle the correct answer.

**9 marks**

|                                                                                                                                                                        |                                                         |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|
| <b>1.</b> If <b>n=6</b> , what will be the result of:<br>switch(n) {<br>case 3: cout << (n-5)<<endl;<br>case 6: cout << (n+5)<<endl;<br>default: cout << (n*5)<<endl;} | a) 11    b) 30<br>30        11<br><br>c)11        d) 30 |
| <b>2.</b> What will be the result of:<br>int n=6;<br>cout<< (3n);                                                                                                      | a) error    b) 36<br>c) 18        d) 63                 |
| <b>3.</b> Is this code correct?<br>char A='R';<br>cout<<R;                                                                                                             | a) Yes<br>b) No                                         |

**b.** Evaluate the following expressions if y=2:

**6 marks**

a)  $Z = 3 * 8 / 2 * (2 * y + 5 / 2);$

b)  $Z = 7 * y \% 2 + 2 * (3 + (y \% 3 + 2));$

c)  $y = ((2 == 4) \&\& (7 > 5));$



Student Name:

Q1/a. Circle the correct answer.

9 marks

|                                                                                                                                                                     |                                          |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------|
| 1. If <b>n=3</b> , what will be the result of:<br>switch(n) {<br>case 3: cout << "DAD ";<br>case 4: cout << "MOM \n"; break;<br>default: cout << "SIS \n"; break; } | a) DAD<br>b) DAD MOM<br>c) MOM<br>d) SIS |
| 2. What will be the result of:<br>cout<< (6++3);                                                                                                                    | a) error    b) 63<br>c) 36       d) 9    |
| 3. Is this code correct?<br>char a="C";<br>cout<< a;                                                                                                                | a)Yes       b)No                         |

b. What is the output of the following code?

4 marks

```
int main()
{ int k= 5, m= 2;
  k -= 4 + ++m;
  m = k + m;
  k-= m;
  cout<<"k = "<<k<<endl;
  cout<<"m = "<<m<<endl; }
```

answer:

c. For the following algebraic expressions, write corrected C++ expressions:

2 marks

1.  $(4.5+5.2\div1.2)-1.4$

2. 
$$\frac{4.12}{12.3-1.6}$$

**Q2:** Write a C++ program to input two integer values that represent two sides of a shape. If the two values are equal, the shape is **square** and its area is calculated. If they are not equal, it is **rectangular** and its area is calculated. **8 marks**

**b.** Rewrite the following C++ program using a switch statement

**7 marks**

| <u>If ... else</u>                                                                                                                                                                                                                                                                     | <u>Switch ... case</u> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| <pre> #include&lt;iostream&gt; using namespace std; int main() { int tem,x;   cout&lt;&lt;"Enter tem: ";   cin&gt;&gt; tem;   if (tem == 1000)     x = 0;   else if (tem&gt;=10 &amp;&amp; tem&lt;1000)     x = 1;   else     x = -1;   cout&lt;&lt;"x ="&lt;&lt;x&lt;&lt;endl; </pre> |                        |